

Conflict and Peace

Week 5: Impact of crime and conflict on individuals

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How do conflicts affect individuals?

Conflicts have very clear short-run implications

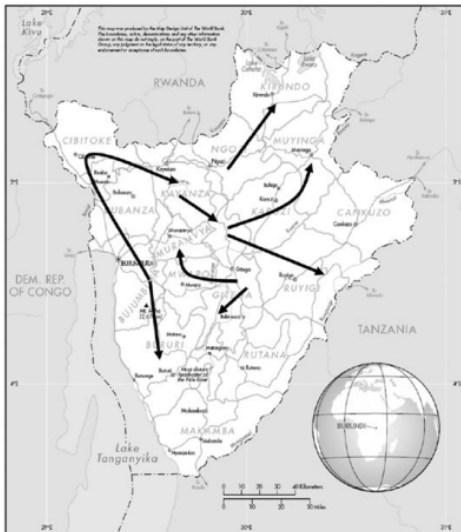
- Loss of life - both in terms of deaths and reduced healthy years
- Destruction of physical capital and infrastructures
- Reduced economic activity and human capital accumulation

What is very worrisome: Negative effects can persist

- How do former combatants reintegrate?
- Does it affect how we value future? (∴ Uncertainty about stable future)
- Does it lock-in vulnerable individuals to crime / conflict specific capital?

Some effects are already long-run!

Negative permanent health effects (Bundervoet et al 2009)



Some effects are already long-run!

Negative permanent health effects (Bundervoet et al 2009)

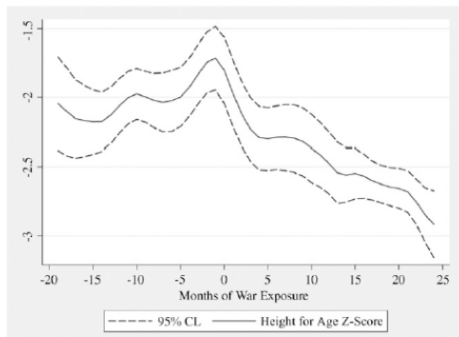


Figure 3

Height for Age Z-Scores, by Months of War Exposure, Treating Nonexposed Children as Having Negative Exposure Equal to a Child's Age

Notes: Kernel-weighted local polynomial regression (using Epanechnikov kernel) of height for age z-score on months of civil war exposure. For those children not exposed to the civil war, months of war exposure takes the negative value of the child's age in months minus six months, so that the youngest nonexposed child is considered to have zero months of war exposure (note that only children older than six months were included in the anthropometrics survey). Dashed lines represent the 95 percent confidence bounds, using a bootstrap simulation with 1,000 repetitions to estimate the standard errors. Data source: World Bank and Burundi Statistics Institute 1998 Priority Survey.

Conflicts can trigger other channels with long-run implications

Interrupting human capital accumulation

- Unable to acquire skills at schools
- Could have implications in the long-run labor market performance
- Potentially may acquire skills only applicable in limited (illicit) sectors

But studying this topic is very tricky

- Data is rare: Mostly developing countries + Dangerous to obtain them
- Need to observe these individuals in the long-run
- Even if acquired, endogeneity issues!
 - ▶ Ensuring that exposed individuals are comparable to the non-exposed are challenging
 - ▶ Measurement error in being exposed to crime / conflict

What we do today

Highlight novel channels that highlight effect of conflict in individuals

- Ability to make long-run investment ↓
 - ▶ Become more risk-averse
 - ▶ Uncertain environments (state presence, underlying policy changes)
 - ▶ Equally affect individuals and firm decisions
- Accumulation of specific capital that does not apply to general skills
 - ▶ Illegal industry specific capital: Difficult to look for other opportunities
 - ▶ Also potentially explain their criminal life path

We highlight potential channels that explain how effects of conflicts in individuals persist

Roadmap for today

1. Overview of the topic

2. Impact on individual and firm decision-making

- 2.1 Brown, Montalava, Thomas, Velásquez (2019): Impact of Violent Crime on Risk Aversion: Evidence from the Mexican Drug War
- 2.2 Blair, Christensen, Wirtschafter (2022): How Does Armed Conflict Shape Investment? Evidence from the Mining Sector
- 2.3 Mirenda, Mocetti, Rizzica (2022): The Economic Effects of Mafia: Firm Level Evidence [Student presentation]

3. Effects on lifepath of individuals

- 3.1 Humphreys, Weinstein (2007): Demobilization and Reintegration
- 3.2 Sviatchi (2022): Making a Narco: Childhood Exposure to Illegal Labor Markets and Criminal Life Paths
- 3.3 Couttenier, Petrencu, Rohner, Thoenig (2019): The Violent Legacy of Conflict: Evidence on Asylum Seekers, Crime, and Public Policy in Switzerland [Student Presentation]

Brown, Montalava, Thomas, Velásquez (2019):
Impact of Violent Crime on Risk Aversion:
Evidence from the Mexican Drug War

Q: How does exposure to crime affect attitudes towards risk

Attitudes towards risk influence key decisions over the course of one's life

- Willingness to take risks affect choices on investment, occupation, migration etc
- Less known: How do life events change willingness to take risks?
 - ▶ Some studies in Psychology dating 1990s (Carmil & Breznitz 1991)
- Causal analysis may be difficult
 - ▶ Exposure to life events that can change attitudes may be selective to begin with
 - ▶ Thus, those exposed to such events may differ on many dimensions to those not exposed

This paper uses panel survey data and variation in crime exposure

- Uses surveys conducted before and after surge of violence following War on Drugs
- Leverages location, timing, and magnitude of violence as variation

Violence in Mexico

Surge of violence in mid-2000s

- Homicide rate < 10 per 100,000 before $\rightarrow \simeq 30$ after (currently 5th highest)
- $\uparrow$ police vs drug cartels, fights among splintered cartels since onset of War on Drugs
- Splintered cartels spread throughout Mexico

How these events could affect attitudes towards risk

- Fear $\uparrow$ $\rightarrow$ Optimism $\downarrow$, Risk-taking $\downarrow$ (Risk-averseness $\uparrow$)
 - Economic conditions $\downarrow$ $\rightarrow$ Risk-taking $\downarrow$ (Risk-averseness $\uparrow$)
 - Health $\downarrow$ $\rightarrow$ Risk-taking $\downarrow$ (Risk-averseness $\uparrow$)
- $\rightarrow$ Goal is to distinguish these channels using data

Data and empirical strategy

Data sources

- Individual panel surveys (Mexican Family Life Survey 2 and 3)
 - ▶ Socioeconomics and demographics
 - ▶ Risk aversion measured with question on types returns from hypothetical gambles
 - ▶ Attrition not correlated with conflict environment and other covariates
- Homicide statistics from government statistics office (INEGI)

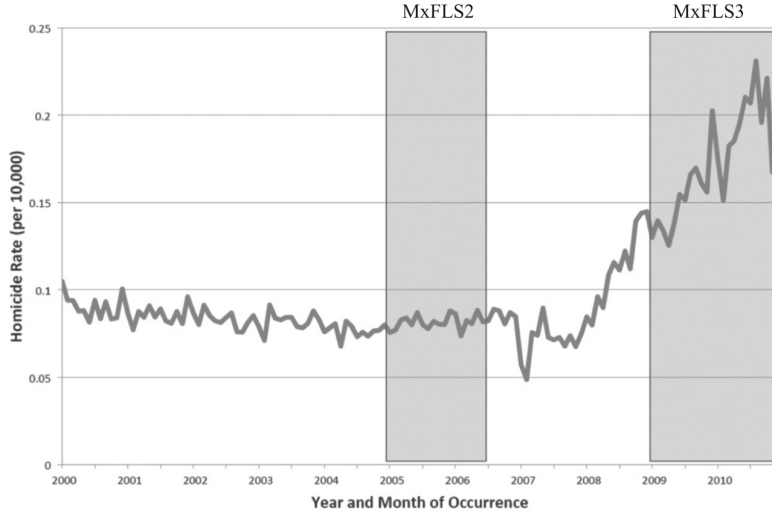
Empirical strategy leverages violence across municipalities and time with panel regression

$$y_{ijmt} = \beta_1 \text{Hom}_{jt} + \beta_2 X_{it-1} + \theta_i + \gamma_t + \lambda_m + e_{ijmt}$$

- $(i, j, m, t) = (\text{indiv}, \text{origin}, \text{destination}, \text{year})$
- $Y_{ijmt} = 1$ if risk averse, X_{it-1} measured year before to avoid reverse causality
- Attrition not correlated with Hom_{jt} (Otherwise, β_1 is biased+-))

Leveraging pre vs post violent waves

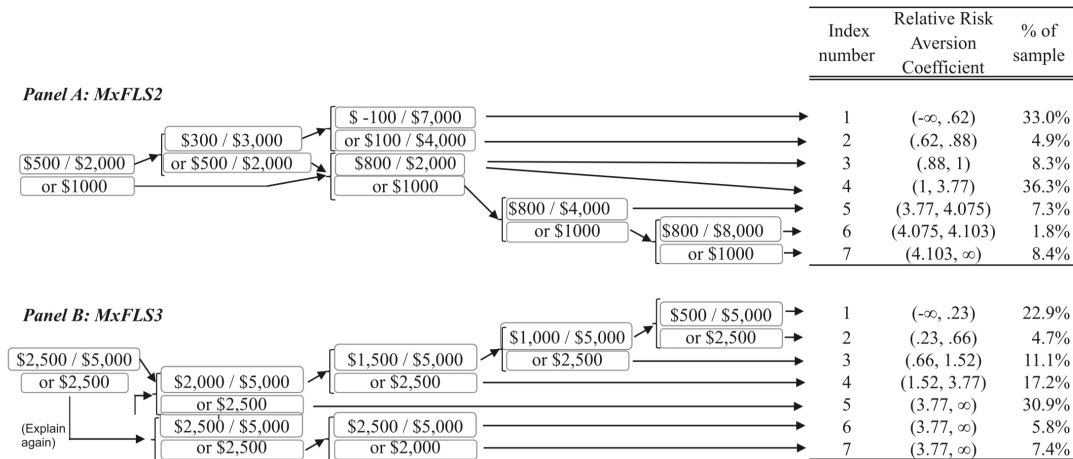
FIGURE 1.—MONTHLY HOMICIDE RATE (PER 10,000)



Data on all reported homicides are collected by the National Institute of Statistics and Geography.

Eliciting risk from survey questions

FIGURE 2.—RISK-AVERSION MEASUREMENT IN MxFLS2 AND MxFLS3



There were 11,348 respondents in each panel. In MxFLS2, risk aversion increases with index number. In MxFLS3, risk aversion increases with index to 5, and gamble-averse respondents are assigned index of 6 or 7.

Cross section vs panel regression discrepancy

TABLE 1.—IMPACT OF VIOLENT CRIME ON RISK AVERSION
(MOST RISK AVERSE = 100)

	Only MxFLS2 (1)	Only MxFLS3 (2)	MxFLS2 and MxFLS3 (3)	MxFLS2 and MxFLS3 (4)
Homicide rate	−2.203*** [0.772]	0.373 [0.329]	1.472*** [0.465]	1.525*** [0.481]
Mean dependent variable	17.51	44.14	30.82	30.82
Observations	11,348	11,348	22,696	22,696
Number of individuals	—	—	11,348	11,348
Individual fixed effects	No	No	Yes	Yes
Municipality fixed effects	No	No	No	Yes

Standard errors clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. All models control for individual characteristics, household characteristics, and date of interview fixed effects.

Why the discrepancy

- MxFLS2: Less risk-averse indivs more likely to stay in high violence areas and agree to be surveyed
- Using panel: Observe same people across time - Able to use fixed effects to control unobservable characteristics

Fear seems to be the most relevant mechanism

TABLE 4.—HETEROGENEOUS IMPACT OF VIOLENT CRIME ON RISK AVERSION BY POTENTIAL MEDIATORS MEASURED IN MxFLS2 AND MxFLS3
(MOST RISK AVERSE = 100)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Homicide rate	1.525*** [0.481]	1.452*** [0.531]	1.453** [0.581]	1.348* [0.743]	2.602 [2.485]	0.447 [0.696]	1.044* [0.591]
Homicide Rate × I(Employed)		0.126 [0.581]					
Homicide Rate × I(Bottom Quartile of Change in PCE)			0.533 [0.650]				
Homicide Rate × Quartic Root of Emotional Well-Being Score				−0.023 [0.493]			
Homicide Rate × Systolic Blood Pressure					−0.010 [0.019]		
Homicide Rate × I(Feel Less Safe Than 5 Years Ago)						2.170** [1.041]	
Homicide Rate × I(Scared of Being Attacked at Night)							1.630* [0.948]
<i>P</i> -value for <i>f</i> -test (Homicide Rate + Homicide Rate Interaction = 0)		0.01	0.00	0.01	0.30	0.00	0.00
Mean dependent variable	30.82	30.81	30.83	30.81	30.76	30.82	30.82
Observations	22,696	22,558	22,212	22,658	16,512	22,540	22,540
Number of individuals	11,348	11,279	11,106	11,329	8,256	11,270	11,270

Standard errors clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. All models control for individual characteristics and household characteristics and date of interview, municipality, and individual fixed effects, as well as the interaction of each of these controls with the relevant subgroup.

Takeaways and discussions

Violence can affect decision-making by affecting risk attitudes

- Violence exposure → Increased risk-averseness (decreased risk-taking)
- Driven by increased sense of fear following violence
- Compares with conventional cross-section estimates: Show that they can be biased

What can we take away from here?

- Consequence of violence: Less willingness to engage in risky but beneficial endeavors
 - ▶ Could lead to lot of missed opportunities for development
 - ▶ Which implies less wealth accumulation and income
- Another channel in which violence hurts development
 - ▶ Violence need not destroy capital directly to impact growth and development

Blair, Christensen, Wirtschafter (2022):
How Does Armed Conflict Shape Investment?
Evidence from the Mining Sector

Q: How do conflicts affect firms' investment decisions?

Uncertainty can disincentivize investment

- Armed conflicts increase uncertainties over future economic returns
- Many measures taken to reduce armed conflict
 - ▶ e.g. Monopolize violence (only the state can legally control military/police etc)
- But do armed conflicts have unilateral effects on investment incentives?

This paper tells us that armed conflicts do have divergent effects on investments

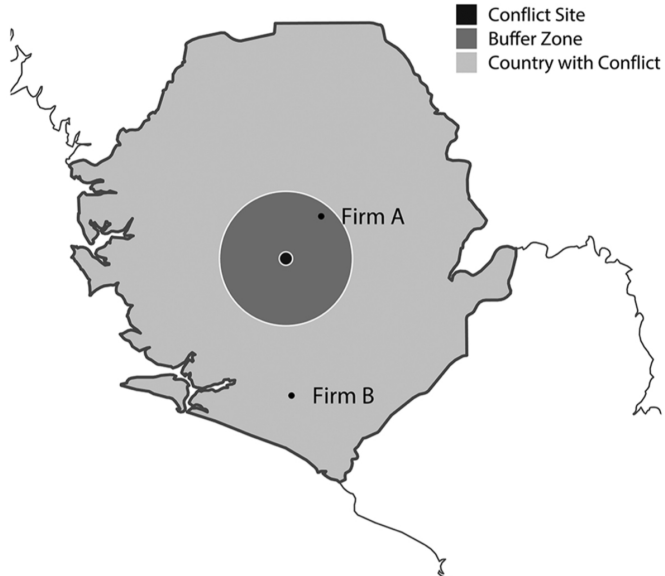
- Even within the same country, distance from conflict may affect changes to incentives
- Destruction of production, state absence, uncertain policies apply differently
 - ▶ Local threats (Destruction, state absence) vs Universal factors (uncertainty)
- Reviews findings from previously published papers

Theoretical framework of conflict and investments

Firms concerned about political instability to the extent that it constrains operations

- Factors to consider: Production closure, state absence, uncertainty, reputation
 - ▶ Production closure (-)
 - ▶ State absence: Lax regulations, difficult to tax (+), less protection of property (-)
 - ▶ Uncertain policies, government action and income horizon (-)
 - ▶ Reputation for working with warring government (-)
- Exposure to political risk is heterogenous (by proximity)
 - ▶ Conflict site: Destruction of production sites → Investment ↓↓
 - ▶ Buffer zone: Ungoverned and difficult to tax → Investment?
 - ▶ Country with Conflict: Still governed normally → Investment?

Theoretical framework of conflict and investments



Theoretical framework of conflict and investments

Table 2. Mechanisms Linking Violence to Investment

Mechanism	Effect Direction	Geographic Scale		
		Conflict Site	Buffer Zone	Country with Conflict
Disrupted production:				
Fighting disrupts operations.	—	✓		
State capacity:				
Tax and regulatory obligations decline in disputed territory.	+	✓	✓	
Ability to protect property rights and maintain critical infrastructure and services declines in disputed territory.	—	✓	✓	
Policy change:				
Conflict increases uncertainty around the standing or actions of government.	—	✓	✓	✓
Reputation:				
Conflict creates risk of reputation loss from investors, home governments, media, or NGOs.	—	✓	✓	✓

Data and empirical strategy

Data sources

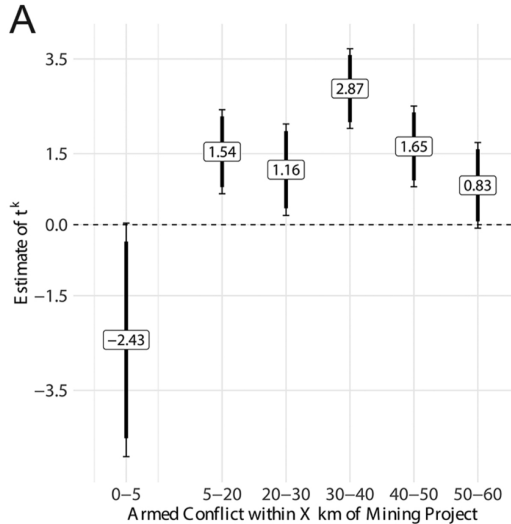
- Investment: Mining investment from SNL Metals and Mining dataset
 - ▶ Includes data on investments made, locations from 1997-2014
- Conflict data from UCDP dataset

Research design leverages variation in distance from conflict site and investment

$$y_{ict} = \alpha_{ic} + \delta_{it} + \theta C_{ct} + \sum_k \kappa^k D_{ict}^k + e_{ict}$$

- firm-country, firm-time fixed effects
- C_{ct} capture whether a conflict happened at country c in year t
- k is a distance bandwidth
- D_{ict}^k captures whether firm i in c is exposed to conflict at year t in bandwidth k

Effects differ by proximity



Yet these heterogeneities are not captured at aggregate level

Table 5. Effect of Armed Conflict on Investment at the Country Level

	Log(Exploration Investment + 1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1 (Conflicts > 0)	-.65** (.32)	-.56 (.34)					
1 (Conflicts = 1)			-.41 (.37)				
1 (Conflicts > 1)			-.74** (.37)				
1 (State-based > 0)				-.17 (.39)			.07 (.39)
1 (One-sided > 0)					-.87*** (.31)		-.85*** (.32)
1 (Nonstate > 0)						-.69 (.45)	-.62 (.46)
<i>p</i> -value on <i>F</i> -statistic	.04	.1	.12	.65	.01	.13	.02
Country-year sample	All	Recipients	All	All	All	All	All
Country FE	177	145	177	177	177	177	177
Year FE	18	18	18	18	18	18	18

Takeaways and discussions

Conflicts do not have unilateral influence on investment outcomes

- Different factors at work depending on proximity to conflict
- Can take advantage of lax rules due to state absence? they may increase investments
- Otherwise, conflicts can discourage investment

Possible extensions

- Effects of state absence on investments (later: Sanchez de la Sierra (2020))
 - ▶ Non-state actors can take over (call them bandits)
 - ▶ If protecting business activity is in bandits' interests, economic activity can continue
 - ▶ Otherwise, they have no interest to protect
- This difference could appear in dimensions other than proximity
 - ▶ Difference by sector of industry, for instance

Mirenda, Mocetti, Rizzica (2022):
The Economic Effects of Mafia: Firm Level
Evidence [Student presentation]

Humphreys, Weinstein (2007): Demobilization and Reintegration

Q: What determines successful demobilization and reintegration?

Many efforts to end protracted conflicts and bring normalcy afterwards

- Peacekeeping operations (PKOs) led by the United Nations
 - ▶ Reform institutions and rebuild infrastructures to reduce incentives for further violence
- Disarmament, demobilization, and reintegration (DDR) is key part of this process
 - ▶ El Salvador, Cambodia, Sierra Leone, Tajikistan etc
- Not enough micro-level evidence to assess the success

This paper uses micro-level evidence to identify factors contributing DDR programs

- Survey of combatants from warring factions in Sierra Leone
- Variation in the degree of reintegration across former combatants
- Identifies factors - personal characteristics, and contexts - that explains DDR effects

Setting: Post-civil war in Sierra Leone (SL)

History of post-war progress

- War declared over in January 2002 by SL government (after 11 years!)
- 5 warring entities involved - War ended with capture of leader of a major faction

DDR process

- Assemble at reception center → Disarm → receive allowances, necessities → civilian
- Receive job training (equally across 5 factions, uncorrelated with political affiliations)

Hypotheses on successes of DDR

- Success depends on dissolving combatant networks, improving earnings, fostering trust on democratic process, and reconciling with families and friends
- Those not happy with earnings potential + distrust other groups less likely to integrate
- Conversely, DDR could be most pronounced on high dissatisfaction and distrust indivs

Data and empirical strategy

Outcome variables from surveying 1000+ combatants across warring groups

- Delinked: Break ties with the faction? (time/business relation/seeking help)
- Employed: Reintegrate to workforce
- Democratic: Confident in democracy? (elect/talk to officials vs fight/traditional leader)
- Accepted: Any problems in gaining acceptance from family and friends?

Key independent variables of integration

- Distrust: Believe in others to comply to peace agreement?
- Dissatisfaction: Faction benefitted or worse-off after agreement?

The paper estimate correlates across various characteristics and above outcomes

- Probit with dummies for factions applied
- Not necessarily causal: Not a randomized procedure in allocating one to DDR
 - ▶ Refer to Blattman and Berman's works we covered few weeks ago

How are the combatants doing in these outcomes?

Table 2
Measures of Reintegration

Variable	N	Mean	SD	Mean CDF	Mean RUF	Difference
Delinked?	1,007	0.86	0.34	0.88	0.84	.04 (1.52)
Employed?	1,037	0.84	0.36	0.90	0.79	.11*** (4.62)
Democratic?	1,009	0.62	0.49	0.61	0.66	−0.05 (1.57)
Accepted?	1,020	0.93	0.25	0.99	0.86	0.13*** (8.50)

Note: CDF = Civil Defense Forces; RUF = Revolutionary United Front. T-statistics in parentheses.
*significant at 10 percent. **significant at 5 percent, ***significant at 1 percent.

Who are more or less likely to reintegrate to begin with?

Table 4
Nonprogram Determinants of Reintegration Success

	(1) Delinked?	(2) Employed?	(3) Democratic?	(4) Accepted?
Distrust	-0.030 [1.59]	-0.020 [0.96]	-0.157 [3.18]***	-0.014 [1.02]
Dissatisfaction	-0.010 [0.51]	-0.044 [2.44]**	0.039 [0.86]	-0.009 [0.83]
Age	0.002 [1.78]*	0.000 [0.28]	0.002 [1.10]	0.000 [0.35]
Male	-0.089 [3.42]***	-0.012 [0.23]	-0.004 [0.06]	-0.002 [0.20]
Mende	0.004 [0.22]	0.014 [0.64]	-0.006 [0.13]	0.014 [1.93]*
Educated	0.005 [0.40]	-0.026 [2.00]**	0.001 [0.06]	-0.001 [0.09]
Poor	-0.001 [0.03]	0.036 [1.57]	0.089 [2.55]**	0.022 [1.92]*
Abducted	-0.000 [0.01]	-0.039 [1.29]	0.193 [4.39]***	-0.000 [0.02]
Political support	-0.055 [2.22]**	0.008 [0.35]	0.180 [3.53]***	-0.028 [2.59]***
Officer	-0.045 [1.40]	-0.021 [0.54]	-0.096 [2.10]**	0.003 [0.20]
Abusiveness	0.009 [0.08]	-0.064 [1.34]	-0.217 [1.51]	-0.087 [2.66]***
District wealth	0.049 [1.40]	-0.102 [1.99]**	-0.109 [2.09]**	0.005 [0.33]
Community suffering	-0.056 [0.43]	-0.065 [0.30]	0.556 [1.94]*	-0.082 [2.87]***

- Distrust and dissatisfaction only explains a small part of reintegration
- Some work in opposite directions (political support for factions)
- High-ranked, wealthier, and educated ones less likely to reintegrate (on some dimensions only)

No systematic evidence of the effect of DDR on reintegration

Table 5
Impact of DDR Program on Reintegration

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Program Variable	Enter DDR				Completed DDR			
Dependent Variable	Delinked?	Employed?	Democratic?	Accepted?	Delinked?	Employed?	Democratic?	Accepted?
DDR	−0.038 [1.09]	0.024 [0.65]	−0.050 [1.65]*	−0.008 [0.69]	−0.011 [0.42]	−0.095 [4.05]***	−0.017 [0.61]	−0.003 [0.34]
Distrust	−0.029 [1.57]	−0.020 [0.96]	−0.157 [3.20]***	−0.013 [0.99]	−0.030 [1.57]	−0.017 [0.80]	−0.157 [3.17]***	−0.013 [1.02]
Dissatisfaction	−0.009 [0.43]	−0.045 [2.51]**	0.039 [0.85]	−0.009 [0.79]	−0.009 [0.48]	−0.037 [2.09]**	0.040 [0.88]	−0.009 [0.82]
Age	0.002 [1.82]*	0.000 [0.30]	0.002 [1.08]	0.000 [0.40]	0.002 [1.80]*	0.001 [0.64]	0.002 [1.12]	0.000 [0.37]
Male	−0.089 [3.44]***	−0.010 [0.19]	−0.015 [0.21]	−0.001 [0.14]	−0.089 [3.43]***	−0.016 [0.30]	−0.005 [0.07]	−0.002 [0.20]
Mende	0.002 [0.08]	0.016 [0.71]	−0.011 [0.26]	0.014 [1.88]*	0.004 [0.19]	0.012 [0.57]	−0.006 [0.15]	0.014 [1.94]*
Educated	0.005 [0.42]	−0.026 [1.98]**	0.002 [0.08]	−0.000 [0.06]	0.005 [0.41]	−0.025 [1.96]*	0.001 [0.06]	−0.001 [0.09]
Poor	0.001 [0.02]	0.036 [1.56]	0.089 [2.59]***	0.023 [2.04]**	−0.000 [0.00]	0.040 [1.74]*	0.091 [2.63]***	0.022 [1.96]**
Abducted	0.002 [0.05]	−0.040 [1.29]	0.196 [4.47]***	0.000 [0.03]	−0.000 [0.01]	−0.033 [1.02]	0.193 [4.38]***	−0.000 [0.02]
Political support	−0.052 [2.08]**	0.007 [0.27]	0.183 [3.66]***	−0.026 [2.51]**	−0.055 [2.18]**	0.014 [0.60]	0.180 [3.57]***	−0.028 [2.57]**

Takeaways and discussions

The observed correlation between DDR and reintegration is weak

- Dissatisfaction and distrust only explains partially
- Some personal characteristics are more robustly correlated
- Participation at DDR does not contribute to reintegration

So is DDR not effective at all? Not so fast...

- Differential factors within DDR could have different effects
 - ▶ Difference in providing short-run money vs long-run opportunities vs cognitive care
 - ▶ See Blattman, Jamison, Sheridan (2017) - Companion of Blattman, Annan (2016)
- Could be different depending on how conflict ended
 - ▶ This one ended with one side losing
 - ▶ Not all conflicts end this way (most end without clear winners)

Sviatchi (2022):
Making a Narco: Childhood Exposure to Illegal
Labor Markets and Criminal Life Paths

Q: Does early exposure to criminal life affect future life paths?

Illegal markets operated by organized criminal groups involve so many individuals

- The 'Kingpins': Pablo Escobar, Arturo Beltran Leyva, Al Capone ..
- But there are also the 'footsoldiers': Mostly from marginalized backgrounds
- So How did they enter the criminal life?
 - ▶ Spot benefit-cost calculus? Persistent factors that lock them into criminal life?

This paper: Early exposure → formation of very specific human capital → lock-in

- Exogenous shocks to illegal markets in Peru from US intervention in Colombia
- Return to illegal act $\uparrow \rightarrow$ Child labor in coca farming $\uparrow$
- Accumulate human capital specific to illegal activities $\rightarrow$ Incarceration $\uparrow$
- Early interventions (cash transfers for education) may offset these effective

Illegal drug markets in Latin America

Coca Production Colombia, Bolivia, Peru: Only places agroecologically suitable

- Change of drug policy in one can spill over, but to one of these countries only!
- As US shifted anti-drug efforts to Colombia, production shifted to Peru (early 2000s)
- Aerial spraying at Colombia hiked prices → Coca production rises in Peru
- Largest producer of cocaine by 2012

Which individuals are more likely to be 'locked-in' to this sector?

- Labor market situation matters when deciding to continue education or work
- Age 11-14: End of primary school → secondary education vs labor market
- Coca price hike has effect on this group: Induce to switch off from education

Data sources

Data sources

- Coca and other agricultural data: Satellite imagery + Agricultural census
- Coca price: UN Office on Drugs and Crime
- 2 Peruvian household surveys: Educational attainment and labor outcomes
- School and incarceration data obtained from administrative data

Empirical strategy leverages price shocks, timing of exposure, and location

- Labor market effects for indiv i at area d at time t in age bracket k :
$$Y_{idt} = \beta_0(PC_t \times \text{Coca}_d) + \sum_k \beta_k \text{Age}_k \times (PC_t \times \text{Coca}_d) + \alpha_d + \phi_t + \sigma_r t + X_{idt} + e_{idt}$$
- Long-run effects (incarceration and crime) for each birth cohort c
$$Y_{dc} = \beta \text{Priceshock} \times \text{Age}_{11-14}_{dc} + \sum_k \beta_k \text{Priceshock} \times \text{Age } k_{dc} + \alpha_d + \delta_c + \sigma_d c + e_{cd}$$
- Why? Incarceration data at cohort level (not individuals)

Geographic and Temporal variation

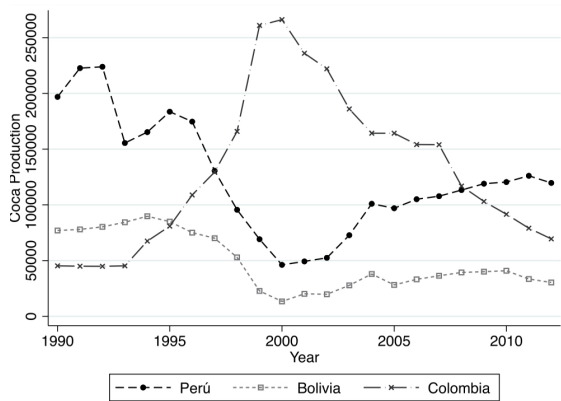
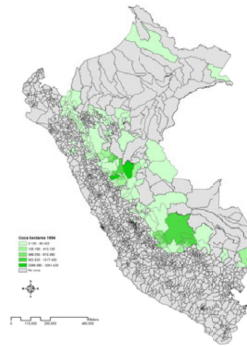
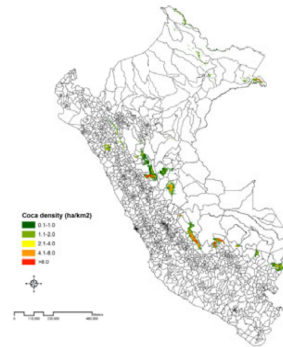


Figure: Coca production over time



(a) Coca production in 1994 across districts



(b) Coca satellite images during the 2000s

Figure: Coca suitability (1994) and production intensity (2000s)

Price shock increase illegal labor participation, esp for 11-14 yr olds

EFFECT OF COCA PRICES ON LABOR PARTICIPATION BY AGES.

	(1) Labor participation	(2) Labor participation
$PriceShock_{d,t}$	0.244 (0.086)	
$PriceShock_{d,t} \times 6x10$	0.043 (0.029)	
$PriceShock_{d,t} \times 11x14$	0.144 (0.028)	0.209 (0.030)
$PriceShock_{d,t} \times 15x18$	0.070 (0.036)	
Total effect at age 11–14	0.388 (0.094)	
Observations	412,026	167,901
Number of districts	1469	1467
Dep. var. mean	0.369	0.299
District FE	✓	✓
Year FE	✓	✓
Household FE		✓

This exposure locks them into criminal industry-specific capital

COCA PRICES DURING CHILDHOOD AND SUBSEQUENT CRIMINAL BEHAVIOR.

	(1) All	(2) Drugs	(3) Violent	(4) Sexual	(5) Family	(6) Other
Panel A: All prisons						
<i>PriceShockAge11to14_{d,c}</i>	3.607 (0.917)	2.127 (0.732)	1.304 (0.384)	-0.107 (0.212)	0.046 (0.088)	0.237 (0.356)
Observations	23,853	23,853	23,853	23,853	23,853	23,853
Dep. var. mean	4.565	0.616	2.269	0.401	0.089	1.190
Panel B: Women						
<i>PriceShockAge11to14_{d,c}</i>	0.436 (0.135)	0.355 (0.150)	0.057 (0.037)	0.001 (0.001)	-0.000 (0.000)	0.024 (0.063)
Observations	23,853	23,853	23,853	23,853	23,853	23,853
Dep. var. mean	1.528	0.623	0.324	0.009	0.001	0.571
Panel C: Legal Coca						
<i>PriceShockAge11to14_{d,c}</i>	4.091 (1.046)	2.542 (0.862)	1.379 (0.405)	-0.305 (0.189)	0.030 (0.105)	0.445 (0.342)
<i>PriceShockAge11to14_{d,c} × Legal</i>	-3.749 (1.675)	-2.654 (1.014)	-0.965 (1.485)	0.994 (0.683)	0.049 (0.149)	-1.173 (1.346)
Observations	23,853	23,853	23,853	23,853	23,853	23,853
Dep. var. mean	4.565	0.616	2.269	0.401	0.089	1.190
District of birth FE	✓	✓	✓	✓	✓	✓
Year of birth FE	✓	✓	✓	✓	✓	✓

Timely educational intervention offsets lock-in effects

CONDITIONAL CASH TRANSFER AND COCA PRICE SHOCKS ON CHILD LABOR.

	(1) Labor	(2) Labor	(3) Labor
$PriceShock_{d,t}$	0.262 (0.104)	0.213 (0.112)	
$PriceShock_{d,t} \times CCT_{d,t}$	0.010 (0.071)	0.024 (0.072)	
$PriceShock_{d,t} \times 6 \times 10$	0.050 (0.043)	0.048 (0.043)	
$PriceShock_{d,t} \times 11 \times 14$	0.163 (0.043)	0.161 (0.043)	0.215 (0.035)
$PriceShock_{d,t} \times 15 \times 18$	0.099 (0.042)	0.098 (0.042)	
$PriceShock_{d,t} \times CCT_{d,t} \times 6 \times 10$	-0.108 (0.064)	-0.107 (0.064)	
$PriceShock_{d,t} \times CCT_{d,t} \times 11 \times 14$	-0.122 (0.048)	-0.121 (0.048)	-0.163 (0.065)
$PriceShock_{d,t} \times CCT_{d,t} \times 15 \times 18$	-0.120 (0.053)	-0.119 (0.053)	
Total effect at age 11–14	0.425 (0.104)	0.374 (0.115)	
Total effect at age 11–14 including CCT	0.313 (0.090)	0.277 (0.099)	
Observations	412,026	401,941	167,901
Number of districts	1469	1469	1467
Dep. var. mean	0.369	0.369	0.299
District FE	✓	✓	✓
Year FE	✓	✓	✓
Department Trend		✓	
Baseline Trend		✓	
Household FE			✓

Takeaways and Discussion

Exposure to illegal markets and crime locks children into criminal life path

- Drop out of school: Cannot accumulate formal human capital
- More likely to be incarcerated later
- Intervention offsetting incentives to join labor early do mitigate some effects

Discussion

- Hints that environment where children grows matters
 - ▶ Active research in economics, education, psychology (Moving to Opportunity etc)
- Exposure at critical timing can tilt incentives, leading to long-run effects
 - ▶ Exposure to adverse events (crime, natural disasters) in utero

Couttenier, Petrencu, Rohner, Thoenig (2019):
The Violent Legacy of Conflict: Evidence on
Asylum Seekers, Crime, and Public Policy in
Switzerland [Student Presentation]

Takeaways and remaining questions

Conflicts affect individuals through many channel

- Direction destruction of physical capital and life opportunities
 - ▶ Not directly covered this week, but indirectly hinted throughout semester
- Affect perspectives on future outlook through changing attitudes towards risk
- May lock-in individuals exposed to crime, especially at critical timing

Remaining questions

- What is the effective rehabilitation and reintegration methods
 - ▶ Not just conflicts, but applies to criminals (Law and economics, labor economics)
- Any institutional setup that mitigate or aggravates impact on individuals? (Next week)